

Multi-Specific Molecule Discovery With The Metaphore Knowledge Base

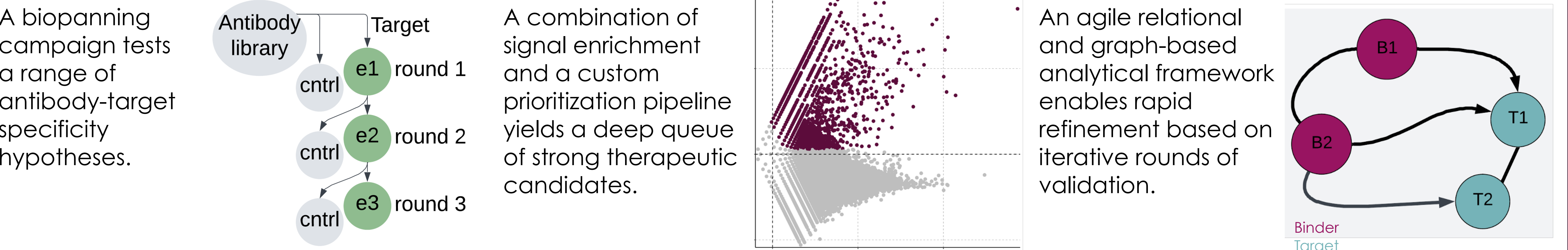
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Multi-specific drug discovery – finding the needle in a haystack, with precision

The identification of a therapeutic antibody agonist capable of acting on multiple targets is a challenging problem, unsolved by the recent advances in high-throughput antibody screening. The Metaphore strategy to reliably pinpoint such multi-specific molecules combines 1) large screens with a range of enrichment conditions with 2) integrated analytical knowledge systems to flexibly nominate molecules for testing. Here, we demonstrate the success of this framework at identifying validated multi-targeting molecules.

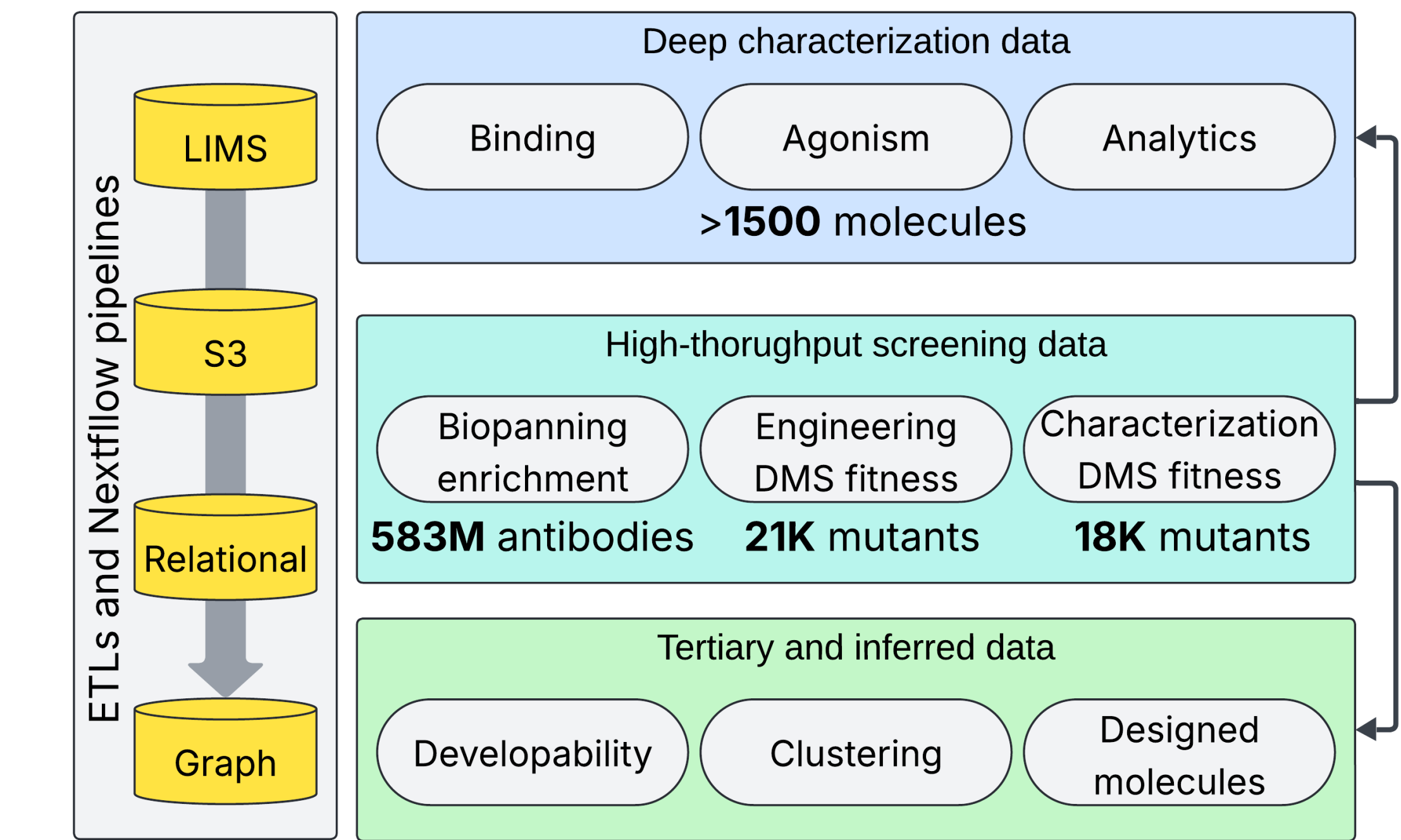
Metaphore antibody prioritization strategy: from high-throughput screening to in-depth validation



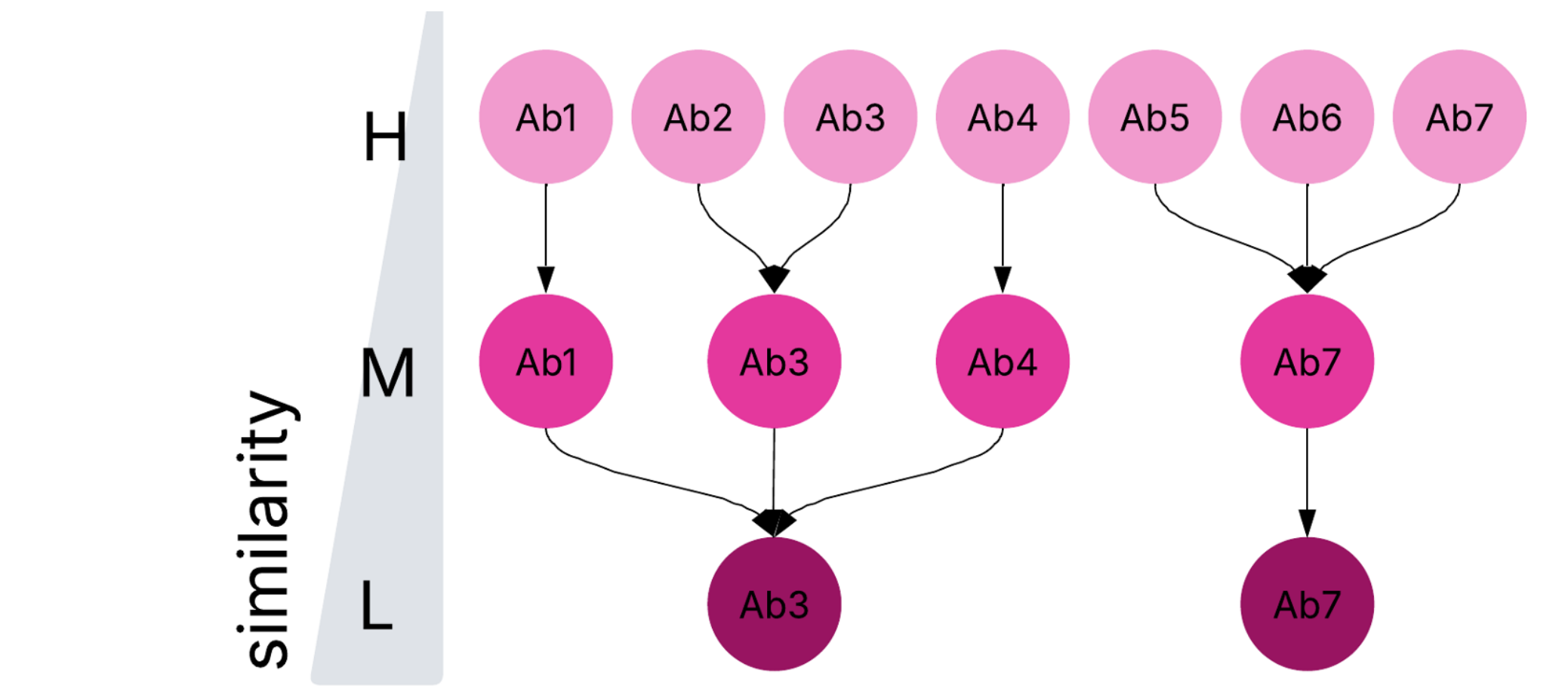
Big data enabled knowledge base enables rapid multi-targeting antibody nomination

1B+ measurements from a range of platforms enabled by a cutting-edge analytical stack

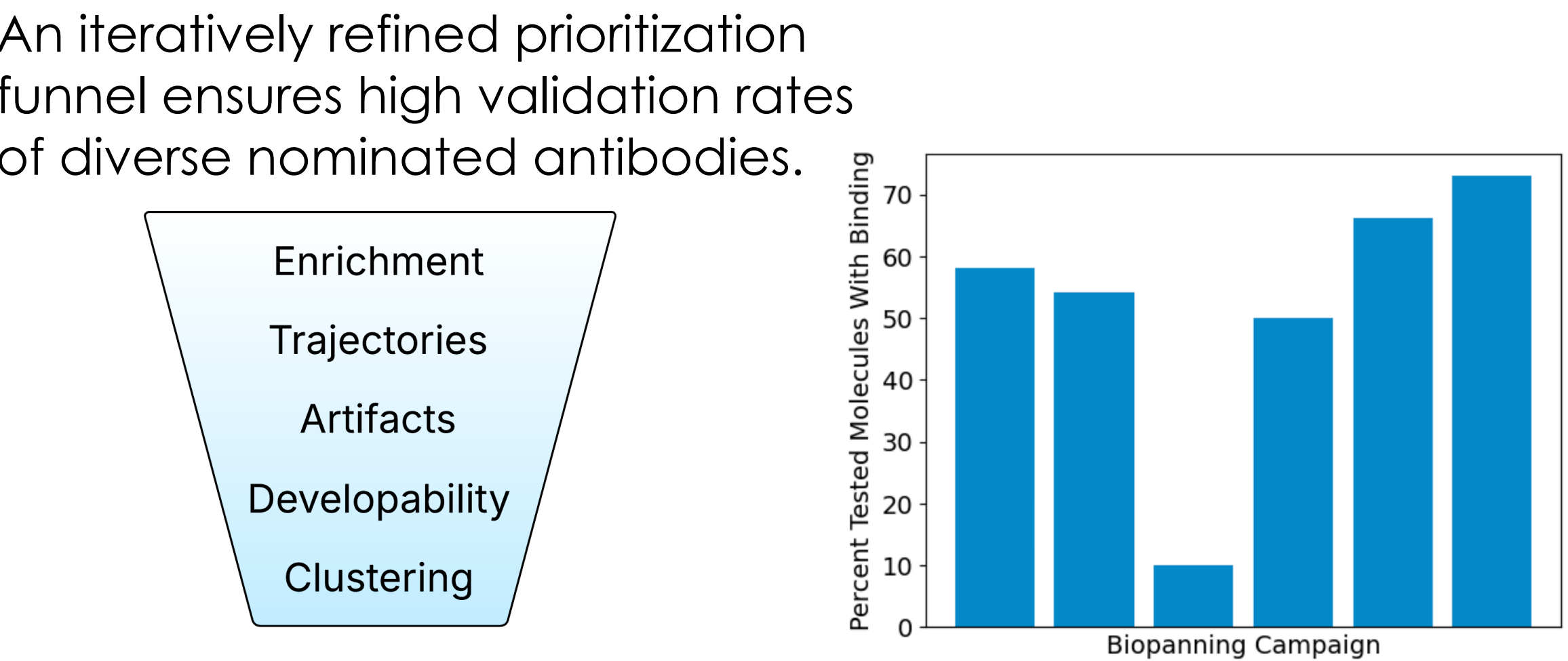
The end-to-end data integration framework incorporates NextFlow-enabled analysis pipelines, relational and graph data solutions, and automated ETLs to enable iterative hypothesis testing and refinement.



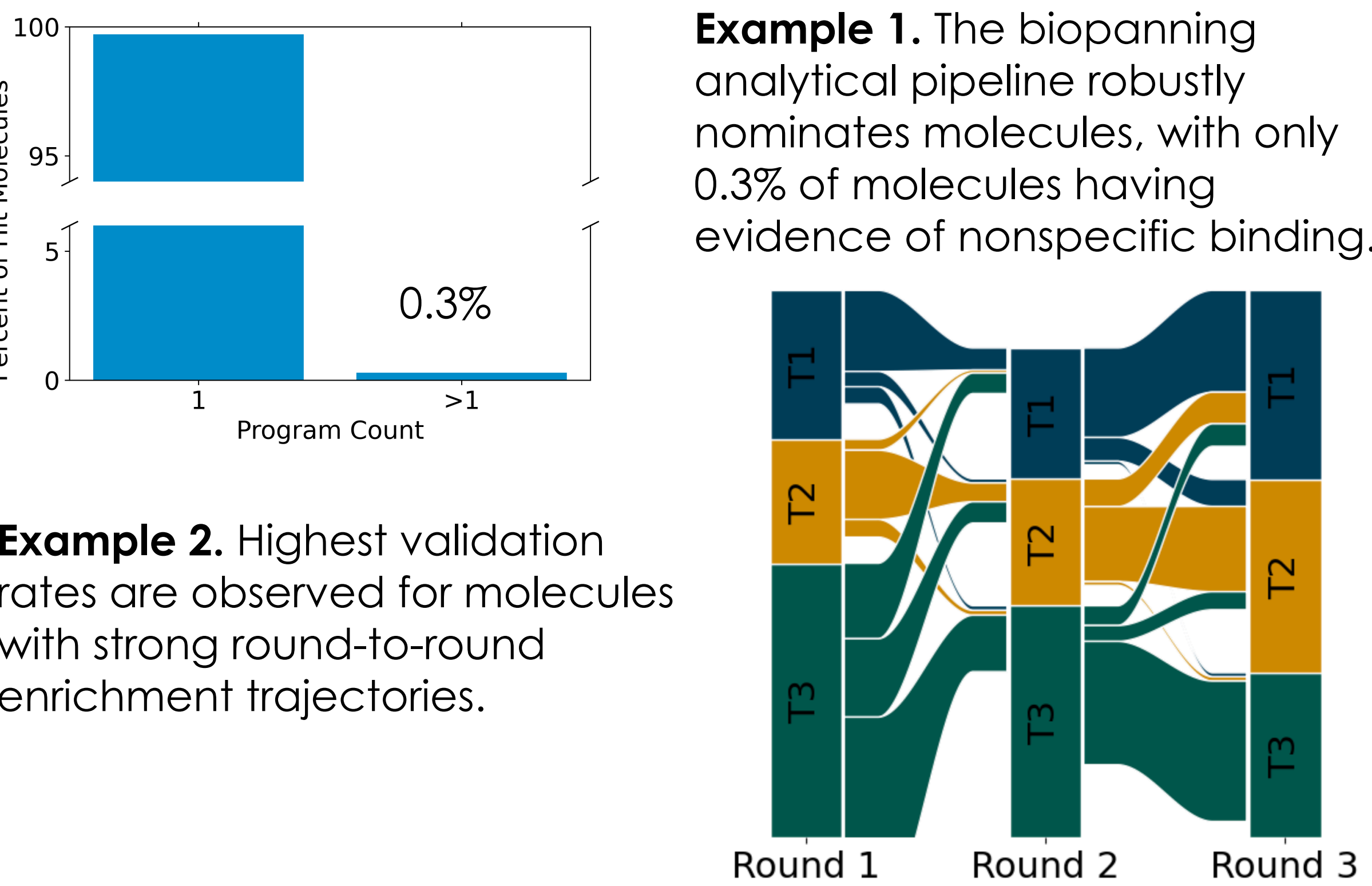
Hierarchical, interpretability-focused MMSeqs2⁽¹⁾ clustering of all 583M antibodies enables the guided exploration of sequences of high and low similarity.



An antibody validation rate >50% routinely achieved for established experimental designs



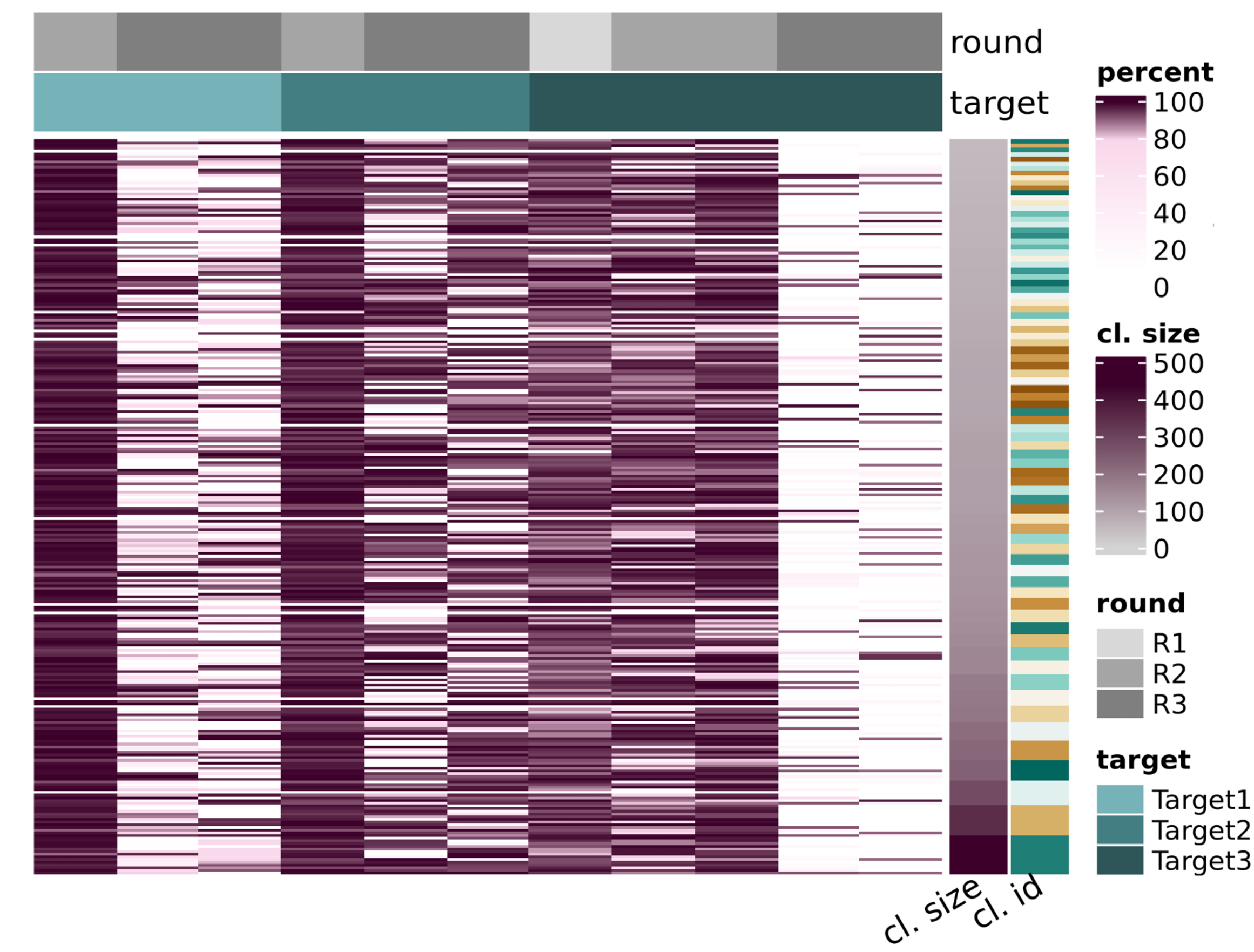
Foundational characterization of molecular metadata improves candidate nomination



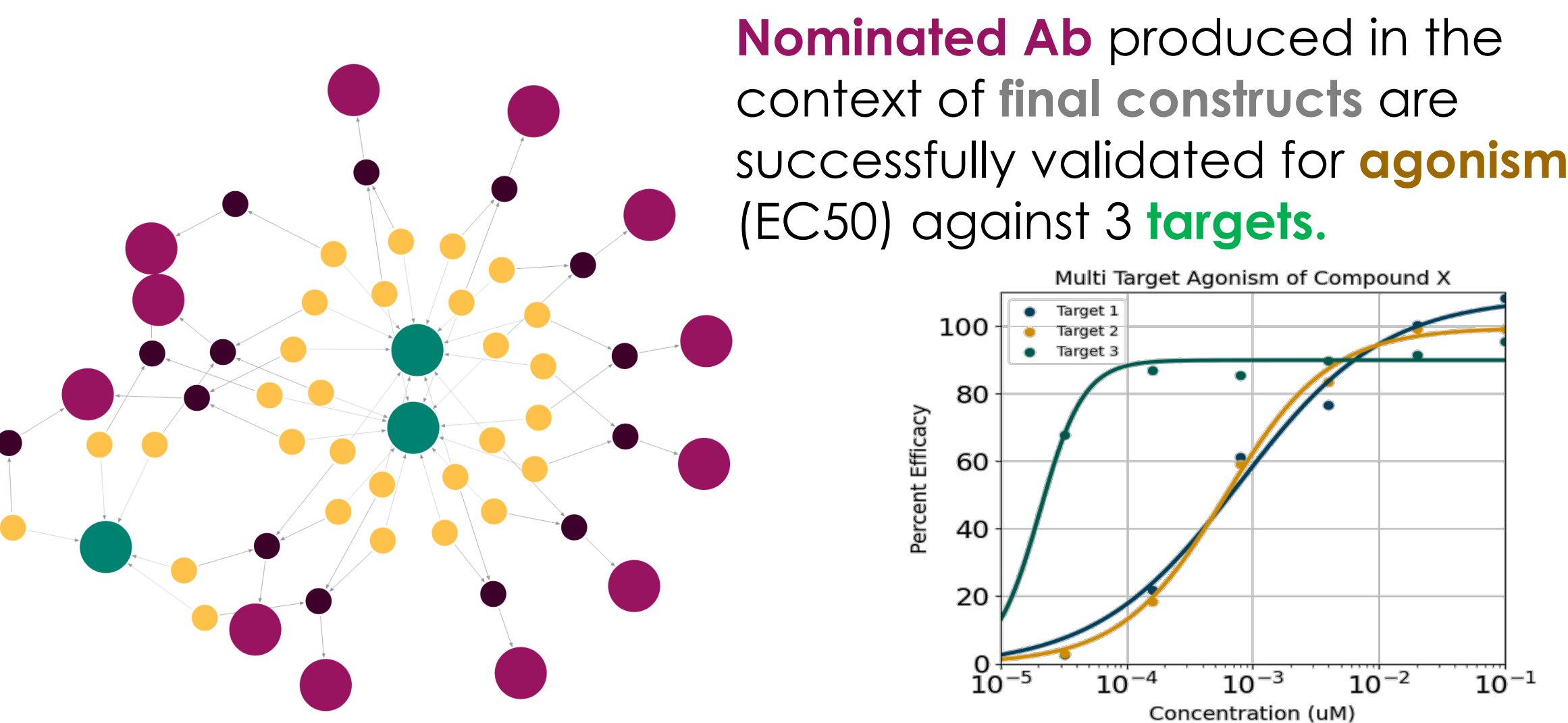
Multi-targeting antibody nomination framework produces validated triple-agonists

Diverse biopanning strategies yield 466K early multi-targeting antibody candidates

Biopanning enrichment (percentile) over multiple rounds (round) and conditions (columns) for 80 select triple-specific (target) antibody families (cl. id), each including at least 50 distinct antibody sequences (rows and cl. size).

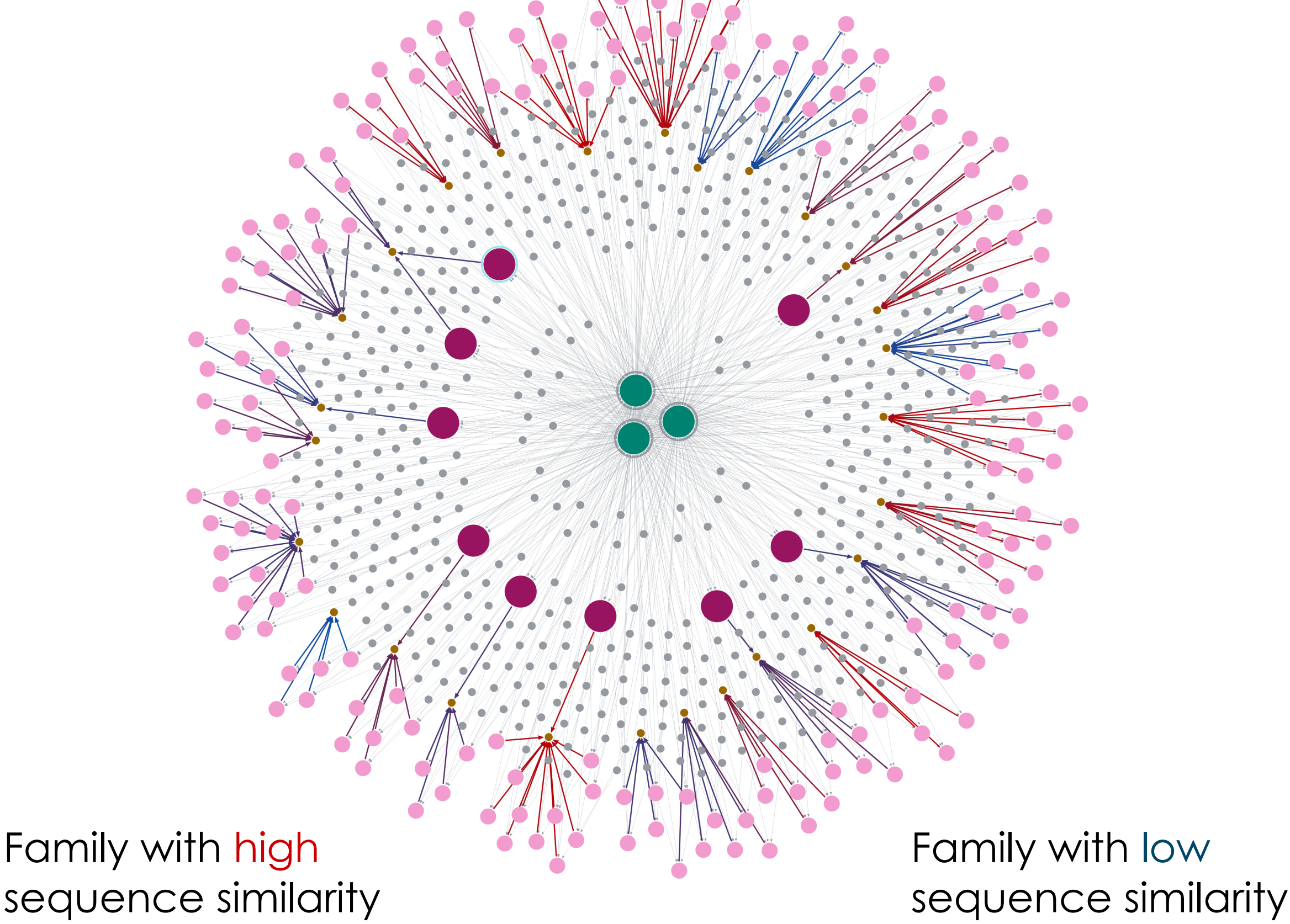


Multi-targeting antibodies validate for agonism in the context of complex molecular modalities

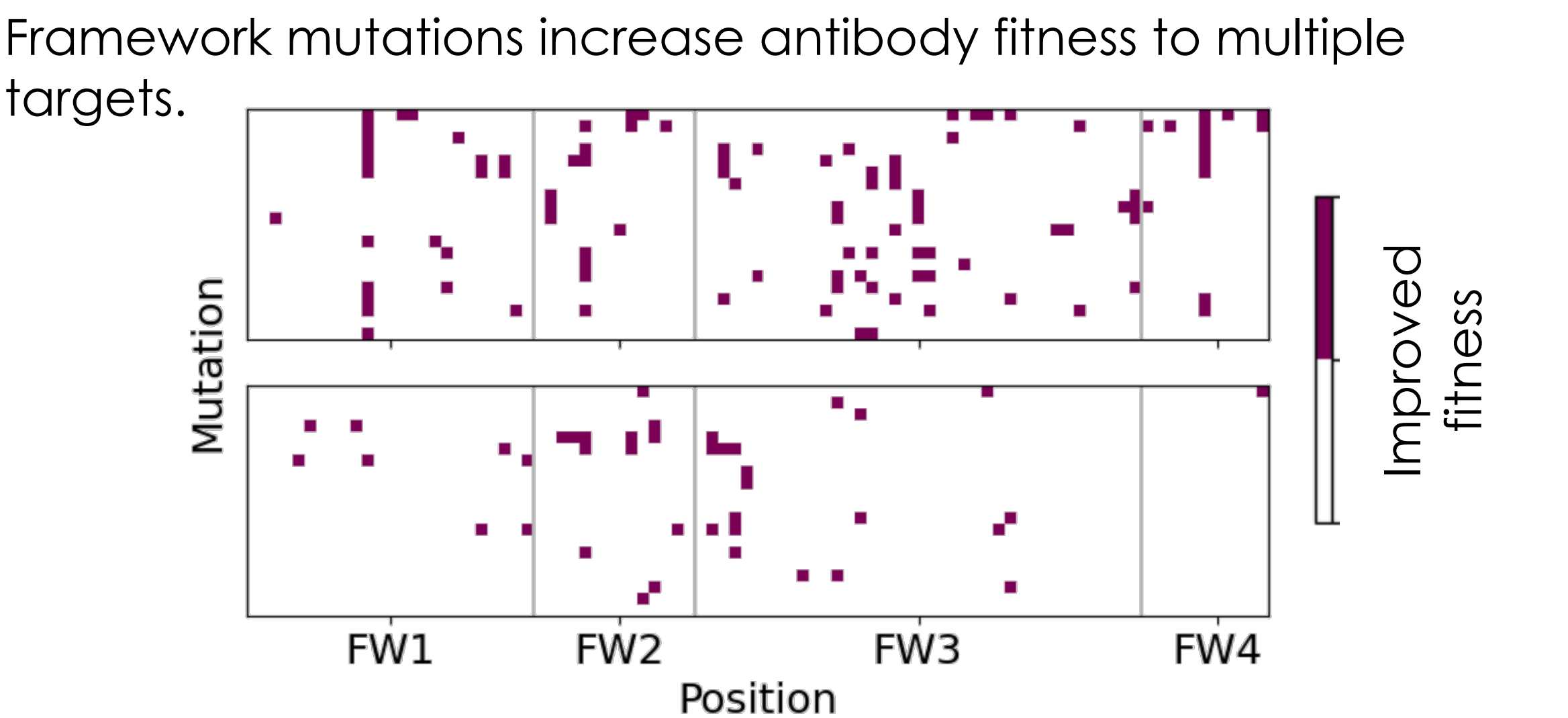


Antibody nomination for validation accounts for sequence diversity and developability properties

Representative 197 antibodies with strongest triple-specific target enrichment form 27 families by sequence similarity. 9 shown antibodies nominated for production, validation, and characterization.



Further validated antibody optimization enabled by deep mutational scanning



Conclusions

- The Metaphore platform robustly identifies multi-targeting therapeutic molecules.
- A combination of high-throughput screening, an end-to-end analytical framework enabling dynamic data exploration (Metaphore Knowledge Base), and comprehensive characterization of molecules results in a deep queue of candidate molecules.
- The continued dataset ingestion, iterative data integration and augmentation, and AI/ML extension of our toolkit ensures continuous improvement and accelerated therapeutic discovery.

(1) Steinegger et al. Nature Biotechnology 2017